DBMS
Database Management System

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Project Proposal

IT Asset & Hardware Maintenance Log System

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1. Summary

Managing IT assets is an essential responsibility for every organization. Computers, laptops, printers, routers, monitors, and networking equipment must be properly tracked to ensure efficient utilization and timely maintenance.

Many small businesses and educational institutions still rely on spreadsheets or paper records to manage their IT equipment. This often leads to missing devices, inaccurate records, delayed maintenance, and poor asset utilization.

Problem Statement

Small IT offices often experience difficulties because:

- Equipment records are maintained manually.
- Asset ownership is difficult to track.
- Maintenance history is not properly recorded.
- Devices needing repair are forgotten.
- Reports take considerable time to prepare.
- Duplicate asset records are common.
- Missing assets cannot be identified quickly.

These issues reduce productivity and increase maintenance costs.

Proposed Solution

The IT Asset & Hardware Maintenance Log System is a simple database-driven application that allows organizations to maintain records of all IT assets, assign equipment to users or departments, monitor maintenance schedules, and update equipment status in real time. The proposed system provides a centralized database that stores information about every IT asset and enables administrators to:

- Register new hardware assets.
- Assign equipment to employees or departments.
- Track current asset status.
- Record maintenance activities.
- Search assets instantly.
- Generate maintenance reports.

2. Project Objectives

The main objectives of this project are:

1. Develop a centralized IT asset database.
2. Track all hardware assets efficiently.
3. Reduce paperwork.
4. Improve maintenance scheduling.
5. Increase equipment accountability.
6. Minimize asset loss.
7. Simplify reporting.
8. Improve overall IT resource management.

3. Scope of the Project

In Scope

The project covers the following categories of hardware:

- Desktop computers
- Laptops
- Printers
- Routers
- Switches
- Monitors
- UPS devices
- Projectors
- Network devices
- Other IT hardware

Out of Scope

The following are explicitly excluded from the current version of the project:

- Software license management
- Automatic hardware detection
- Barcode scanning
- Mobile application
- Cloud synchronization

4. Methodology

The system will be developed following a modular, database-first approach, moving from data design through functional implementation to testing and reporting.

System Modules

Module 1 – Login System

Provides secure administrator authentication before any asset or maintenance data can be accessed or modified.

Module 2 – Dashboard

Displays summary statistics: Total Assets, Operational Devices, Under Repair, and Maintenance Required.

Module 3 – Asset Management

Provides full CRUD (Create, Read, Update, Delete) operations on asset records.

Module 4 – Maintenance Management

Handles maintenance logs and repair history for each asset.

Module 5 – Reports

Generates printable, categorized reports for administrators and department heads.

Database Design

The system is supported by a relational database consisting of three core tables:

Field	Type	Description
UserID	INT (PK)	Unique identifier for each user
FullName	VARCHAR(100)	Full name of the employee
Department	VARCHAR(50)	Department the user belongs to
Email	VARCHAR(100)	Contact email address

Table: Assets

Field	Type	Description
AssetID	INT (PK)	Unique identifier for each asset
SerialNo	VARCHAR(100)	Manufacturer serial number
DeviceType	VARCHAR(50)	Category of hardware
Brand	VARCHAR(50)	Manufacturer / brand name
Model	VARCHAR(50)	Model number or name

PurchaseDate	DATE	Date of purchase
WarrantyExpiry	DATE	Warranty expiration date
AssignedUserID	INT (FK)	References Users.UserID
Status	VARCHAR(30)	Operational / Maintenance Required / Under Repair / Retired

Table: Maintenance_Log

Field	Type	Description
MaintenanceID	INT (PK)	Unique identifier for each maintenance record
AssetID	INT (FK)	References Assets.AssetID
MaintenanceDate	DATE	Date maintenance was initiated
Technician	VARCHAR(100)	Name of technician performing the work
Problem	TEXT	Description of the reported issue
RepairDetails	TEXT	Description of repair actions taken
Cost	DECIMAL	Cost of the repair
CompletionDate	DATE	Date maintenance was completed

Technology Stack

Component	Technology
Frontend	HTML5
Styling	CSS3
Programming (Client-side)	JavaScript
Backend	PHP
Database	MySQL
Web Server	XAMPP
IDE	Visual Studio Code

Testing Plan

The following test cases will be used to validate core system functionality prior to deployment:

Test Case	Expected Result
Add Asset	Asset saved successfully
Edit Asset	Information updated
Delete Asset	Asset removed
Search Asset	Correct record displayed
Update Status	Status changes successfully
Add Maintenance	Maintenance record stored
Generate Report	Report generated correctly

5. Timeline

The project is scheduled for a total duration of 6 – 8 weeks, as stated in the project overview. A suggested phase breakdown based on the system modules and functional scope defined in this proposal is as follows:

Week(s)	Phase	Key Activities
Week 1	Requirements & Database Design	Finalize functional/non-functional requirements; design ERD and table schemas
Week 2 – 3	Core Development	Build Login, Dashboard, and Asset Management modules (CRUD)
Week 4 – 5	Maintenance Module & Reports	Implement Maintenance Log tracking and Reports module
Week 6	Search, Filter & UI Polish	Implement search/filter functionality; refine dashboard and pages
Week 7	Testing	Execute test cases (Add/Edit/Delete/Search/Status/Report generation)
Week 8	Deployment & Documentation	Final deployment, documentation, and submission

Note: The source proposal specifies only the overall 6–8 week duration; the phase breakdown above is a reasonable allocation inferred from the system modules and testing plan described in the document, not an explicit schedule from the original text.

6. Budget Estimate

The source proposal does not include a budget or cost estimate — the technology stack (HTML5, CSS3, JavaScript, PHP, MySQL, XAMPP, and Visual Studio Code) consists entirely of free and open-source tools, and the project is scoped as an academic DBMS exercise. The table below reflects that: all core components are $0, since no paid infrastructure, licensing, or hosting was specified in the original document.

Item	Notes	Estimated Cost
Frontend (HTML5/CSS3/JS)	Open-source, no license cost	$0
Backend (PHP)	Open-source	$0
Database (MySQL)	Open-source (via XAMPP)	$0
Web Server (XAMPP)	Free local development stack	$0
IDE (Visual Studio Code)	Free	$0
Hosting / Cloud	Not specified — cloud sync explicitly out of scope	Not applicable
Total Estimated Cost		**$0**

7. Risk Assessment

The source proposal does not include a formal risk register. The risks below are inferred from the stated Problem Statement, Scope, and Non-Functional Requirements sections of the original document.

Risk	Likelihood	Impact	Mitigation (derived from proposal)
Duplicate asset records	Medium	Medium	Enforced by AssetID/SerialNo as unique fields in the Assets table
Unauthorized data access	Medium	High	Admin login and secure database connection (Section 7.2, Security)
Data loss / integrity issues	Low	High	Data validation and reliability requirements (Section 7.3)
Missed maintenance tracking	Medium	Medium	Maintenance_Log table with mandatory technician, date, and cost fields
Limited availability outside office hours	High	Low	Acknowledged directly as a system constraint (Section 7.5, Availability)

Scope creep (e.g., barcode/mobile requests)	Medium	Medium	Explicitly listed as Out of Scope / Future Enhancements

8. Stakeholder

Based on the system's Users and Departments structure and its login/role design described in the proposal, the following stakeholders are involved:

Stakeholder	Role	Interest / Involvement
Administrator	System operator	Registers assets, manages maintenance logs, generates reports (Module 1–5)
Employees / Users	Asset holders	Assigned equipment tracked via the Users table (FullName, Department, Email)
Department Heads	Report recipients	Receive department-wise asset and maintenance reports
Technicians	Maintenance staff	Recorded against each maintenance entry (Technician field, Maintenance_Log)
Course Instructor / Evaluator	Academic assessor	Evaluates the project as a DBMS course submission

9. Expected Outcomes

After completing this project, the following outcomes are expected:

- All IT assets will be centrally managed.
- Asset assignment becomes transparent.
- Maintenance history is preserved.
- Equipment downtime is reduced.
- Reports are generated quickly.
- Data accuracy is improved.

Future Enhancements

Future versions of the system may include:

- QR Code integration
- Barcode scanning
- Email maintenance reminders
- SMS notifications
- Mobile application

- Cloud database
- Multi-user authentication
- Role-based access control
- Dashboard analytics
- Asset depreciation calculation
- Automatic warranty alerts
- REST API integration

Conclusion

The IT Asset & Hardware Maintenance Log System is a practical and efficient database management project designed to address the common challenges faced by small IT offices in tracking hardware assets and maintenance activities. By replacing manual record-keeping with a centralized digital system, the project improves accountability, streamlines maintenance workflows, and enables quick access to accurate asset information.

Its straightforward database structure, essential CRUD operations, and reporting capabilities make it an excellent beginner-to-intermediate DBMS project, while also providing real-world value and a strong addition to a software development or IT support portfolio.